

BARKING SANDS, HI, USAWMO: **911620**Lat: **22.033N**Lon: **159.783W**Elev: **7**StdP: **101.24**Time Zone: **-10.00 (W10)**Period: **96-19**WBAN: **22501****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.8	17.6	7.8	6.6	21.5	9.9	7.6	20.7	12.1	23.3	11.3	23.5	2.0	70	0.295

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.1	31.2	23.5	30.7	23.3	30.2	23.0	25.5	29.0	25.0	28.8	24.6	28.5	4.8	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.4	19.4	27.4	23.7	18.6	27.4	23.1	17.9	27.4	77.9	29.2	75.7	28.8	73.9	28.4	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.5	8.5	7.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.6	22.4	22.5	22.7	23.7	24.6	25.5	26.3	26.6	26.6	26.0	24.6	23.3
	DBStd	1.91	1.21	1.30	1.44	1.22	1.28	0.94	0.82	0.79	0.87	0.86	1.05	1.14
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5317	383	349	392	410	452	465	506	516	498	495	438	413
	CDD18.3	2276	125	116	134	160	194	215	248	257	248	237	188	155
	CDH23.3	17517	543	501	674	976	1463	1850	2387	2563	2384	2048	1303	823
	CDH26.7	4127	16	22	53	116	277	460	722	828	771	566	240	56
Wind	WSAvg	2.7	3.3	3.7	3.3	2.7	2.7	2.3	2.4	2.3	2.4	2.4	2.6	2.8
Precipitation	PrecAvg	1500	164	131	169	112	96	75	83	93	88	127	173	194
	PrecMax	2117	418	236	785	244	234	155	121	159	192	219	423	479
	PrecMin	951	39	26	45	57	39	28	62	49	54	51	78	38
	PrecStd	307	94	62	153	48	52	31	16	27	40	52	95	121
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.9	28.8	29.3	29.3	29.9	30.7	31.5	31.8	32.0	31.4	30.1	28.6
		MCWB	21.5	20.1	20.2	21.2	22.4	22.2	23.1	24.2	24.3	22.9	22.4	21.4
	2%	DB	27.1	27.4	27.9	28.5	29.3	30.0	30.7	31.0	31.1	30.5	29.2	27.9
		MCWB	21.2	20.8	20.3	21.3	22.1	22.1	23.1	23.7	23.9	22.9	22.1	21.5
	5%	DB	26.6	26.8	27.1	27.9	28.8	29.5	30.2	30.4	30.5	29.8	28.6	27.3
		MCWB	20.9	20.5	20.2	21.1	22.0	22.0	23.0	23.3	23.6	22.9	21.9	21.5
	10%	DB	26.0	26.1	26.5	27.3	28.3	29.0	29.7	29.9	29.9	29.2	28.1	26.8
		MCWB	20.6	20.2	20.1	20.8	21.8	22.0	22.9	22.9	23.4	22.8	21.7	21.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.7	23.8	23.2	24.1	25.2	24.1	25.4	25.8	26.3	25.7	25.1	24.9
		MCDB	26.0	26.0	26.4	26.3	27.8	27.5	29.2	29.8	29.0	28.4	27.5	26.5
	2%	WB	22.7	22.7	22.2	22.9	24.2	23.6	24.8	25.1	25.5	24.9	24.2	23.4
		MCDB	25.6	25.3	25.3	26.1	27.6	27.5	29.0	29.5	29.0	28.5	27.5	26.3
	5%	WB	21.9	21.9	21.6	22.3	23.2	23.0	24.3	24.6	24.9	24.3	23.5	22.8
		MCDB	25.2	25.0	25.0	25.8	27.1	27.6	28.7	29.1	28.9	28.2	27.2	26.1
	10%	WB	21.4	21.4	21.0	21.7	22.5	22.7	23.8	24.1	24.4	23.8	22.8	22.2
		MCDB	24.9	24.8	24.9	25.7	26.8	27.5	28.5	28.6	28.7	28.0	26.8	25.7
Mean Daily Temperature Range	5% DB	MDBR	7.4	7.2	7.4	7.3	7.2	7.3	7.3	7.1	7.2	7.2	7.4	7.1
		MCDBR	7.7	7.8	8.1	7.9	7.6	7.7	7.6	7.5	7.6	7.6	7.7	7.5
		MCWBR	4.7	4.9	4.6	4.3	4.5	4.2	4.4	4.5	4.5	4.6	4.8	4.8
	5% WB	MCDBR	7.0	6.5	6.9	6.8	7.0	7.1	7.0	6.8	6.7	6.9	7.0	6.5
		MCWBR	4.9	4.6	4.2	4.3	4.5	4.3	4.2	4.1	4.1	4.8	5.1	4.9
Clear-Sky Solar Irradiance	taub	0.354	0.357	0.363	0.396	0.387	0.375	0.375	0.373	0.366	0.369	0.361	0.349	
	taud	2.603	2.588	2.577	2.469	2.523	2.565	2.573	2.578	2.603	2.610	2.626	2.646	
	Ebn at Noon	919	937	941	908	906	909	909	916	924	912	905	914	
	Edh at Noon	89	96	101	114	107	102	101	101	97	92	87	83	
All-Sky Solar Radiation	RadAvg	4.42	4.93	5.72	6.53	7.00	7.30	7.21	6.93	6.44	5.34	4.49	4.05	
	RadStd	0.18	0.42	0.60	0.38	0.40	0.19	0.19	0.26	0.19	0.21	0.15	0.30	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page